

Q1) Fill in table below the correct answer

1- The binary code of $(21.125)_{10}$ is :

- A- 10101.001 B- 10100.001 C- 10101.010 D- 10100.111

2- One of the following is a universal gate:

- A- AND B- NAND C- OR D- XOR

3- The number 1000 would appear just immediately after:

- A- $FFFF_{16}$ B- 1111_2 C- 7777_8 D- All of above

4- Which of the following is a two level-logic gate:

- A- OR B- AND C- XOR D- NOT gate

5- Among the logic families, the family which is used in a systems required high speed operation is:

- A- TTL B- CMOS C- MOS D- ECL

6- Excess-3 is known as :

- A- Weighted code B- Cyclic redundancy code C- Self-complementing code D- Algebraic code

7- Indicate which of the following three binary addition is correct:

- i. $1011 + 1010 = 10101$ ii. $1010 + 1101 = 10111$ iii. $1010 + 1101 = 11001$

- A- i and ii B- i and iii C- iii only D- ii and iii

8- Identify the logic function performed by the circuit shown in the given figure



- A- $\bar{A} + \bar{B}$ B- $\bar{A} + B$ C- $A \oplus B$ D- $\bar{A} \oplus B$

9- The Tri-State Buffer is:

- A- A combinational circuit B- Sequential circuit C- Memory circuit D- B and C

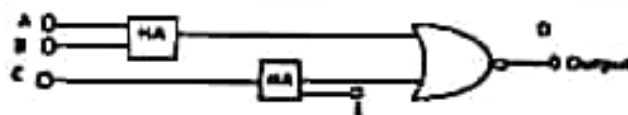
10- What is the minimum number of 2 input NAND gates required to implement the function $(\bar{x} + \bar{y})(z + w)$:

- A- 3 B- 4 C- 5 D- 6

11- If $A \oplus B = C$, then:

- A- $A \oplus C = B$ B- $A \oplus B \oplus C = 0$ C- $B \oplus C = A$ D- A and C

12- The circuit shown below is:



- A- Full Adder B- Full Subtractor C- Parity Checker D- Multiplier

13- Output of the following circuit



is:

- A- 0 B- 1 C- X D- $\bar{X}$

14- The number of 2 input multiplexers required to construct a 2^{10} input multiplexer is:

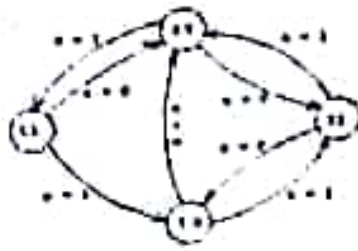
- A- 31 B- 10 C- 127 D- 1023

15- Extremely low power dissipation and low cost can be achieved by:

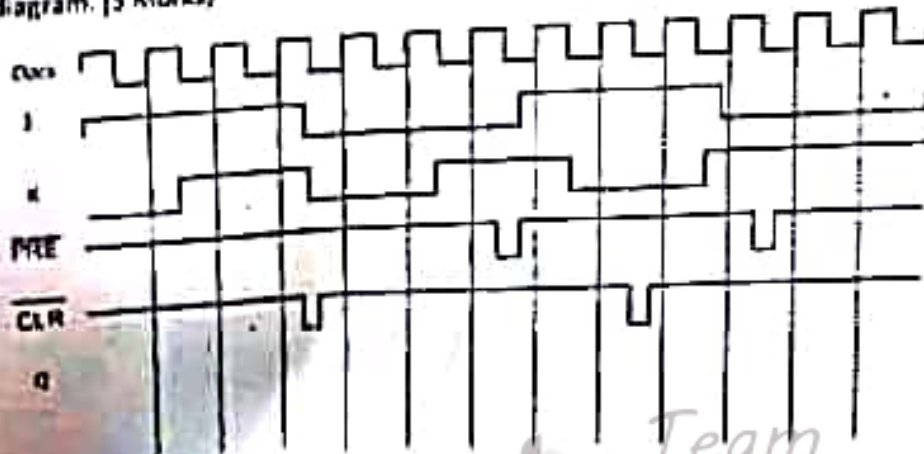
- A- MOS B- CMOS C- TTL D- ECL

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	

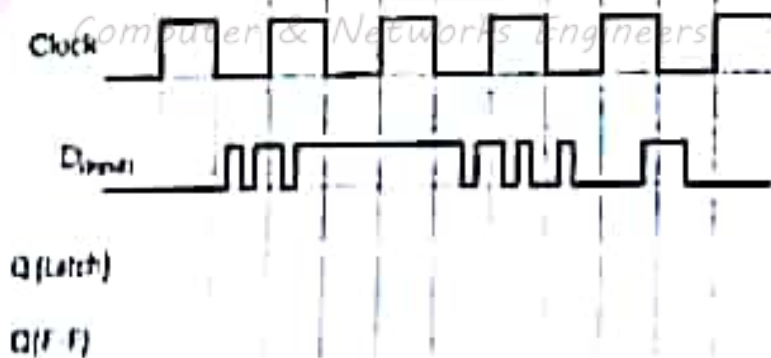
(22) Draw the circuit which implement the state diagram below using J-K flip flop (10 marks)



Q11A For a positive edge triggered J-K flip flop with preset and clear inputs. Determine Q for the inputs shown by the timing diagram. (5 Marks)



Q11B Complete the following timing diagram for a positive trigger 1- D Latch 2- D Flip Flop (5 Marks)



Q4) Using only three state buffer(s) and inverter(s), construct a 4-1 MUX with two select lines. (5 Marks)

Q5) Design asynchronous down counter which counts from 15 to 9. (5 Marks)

Q6) Design a 4 to 2 Priority Decoder such that the inputs are D_3, D_2, D_1, D_0 . D_3 is the most priority (5 marks)

D_3	D_2	D_1	D_0	X	Y	V

Student No.:
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Engineering Technology
Computer Engineering Department
Digital Logic Design Laboratory
Final Exam



1st Semester 2017/2018
Time: One hour

Q1: Choose the correct answer: (40 marks)

- 1) Logic "ZERO" is represented practically by:
A. Vcc
B. GND
C. $\pm 5V$
D. None of the above
- 2) Determine the values of A, B, C, and D that make the sum term $A' + B + C' + D = 0$:
A. $A = 1, B = 0, C = 0, D = 0$
B. $A = 1, B = 0, C = 1, D = 0$
C. $A = 0, B = 1, C = 0, D = 0$
D. $A = 1, B = 0, C = 1, D = 1$
- 3) Output will be "LOW" for any case when one or more inputs are zero for a(n):
A. OR gate
B. NOT gate
C. AND gate
D. NOR gate
- 4) Applying DeMorgan's theorem to the expression $\overline{(x + y) + z}$, we get:
A. $(x + y)z$
B. $(x' + y')z$
C. $(x + y)z'$
D. $(x' + y')z'$
- 5) How many 4-bit parallel adders would be required to add two 7-bit binary numbers?
A. 1
B. 2
C. 3
D. 4
- 6) Convert the following SOP expression to an equivalent POS expression.
 $ABC + AB'C' + AB'C + ABC' + A'B'C$
A. $(A' + B' + C')(A + B + C')(A' + B + C)$
B. $(A + B + C)(A + B' + C)(A + B' + C')$
C. $(A' + B' + C')(A + B' + C)(A + B + C)$
D. $(A + B + C)(A' + B + C')(A + B' + C)$
- 7) When reading a Boolean expression what does the word "NOT" indicate:
A. The same as
B. High
C. Inversion
D. Low
- 8) In half adder EX-OR gate output is:
A. Carry
B. Sum
C. Borrow
D. None of the above
- 9) Two 4-bit binary numbers (1011 and 1111) are applied to a 4-bit parallel adder. The carry input is "0". What are the values for the sum and carry output?
A. Sum = 0111, $C_{out} = 0$
B. Sum = 1111, $C_{out} = 1$
C. Sum = 1010, $C_{out} = 1$
D. Sum = 1100, $C_{out} = 1$
- 10) How many data select lines are required for selecting 64 inputs?
A. 3
B. 4
C. 5
D. 6
- 11) Which of the following could be a canonical representation for $F(A, B, C, D)$:
A. $A'BC + AB'CD + AC$
B. $A'BC'D + ABCD + AB'C'D'$
C. $ABCD + A'B$
D. $A'D + BC' + BD$

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- 12) The devices that convert information from one format to another are :
 A. Encoders and Decoders
 B. Multiplexers and Demultiplexers
 C. Adders and Subtractors
 D. All of the Above
- 13) To implement the Boolean Function $F = A \oplus B \oplus C$ we can use a 3-to-8 decoder and connect:
 A. 3 different lines from its output to an OR gate
 B. 4 different lines from its output to an OR gate
 C. 5 different lines from its output to an OR gate
 D. 7 different lines from its output to an OR gate
- 14) The series of 74xx ICs belongs to the logic family:
 A. TTL
 B. CMOS
 C. ECL
 D. NMOS
- 15) How many 1-of-16 decoders are required for decoding a 7-bit binary number?
 A. 5
 B. 6
 C. 7
 D. 8
- 16) The outputs of Active-low decoders correspond to the:
 A. Minterms
 B. Maxterms
 C. Both A and B
 D. None of the above
- 17) In the given 4-to-1 multiplexer, if $c1 = 0$ and $c0 = 1$ then the output M is:
 A. $X0$
 B. $X1$
 C. $X2$
 D. $X3$
- 18) An example of a combinational circuit is:
 A. Decoder
 B. Multiplexer
 C. Adder
 D. All of the above
- 19) The binary numbers $A = 1100$ and $B = 1001$ are applied to the inputs of a comparator. What are the output levels?
 A. $A > B = 1, A < B = 0, A = B = 1$
 B. $A > B = 0, A < B = 1, A = B = 0$
 C. $A > B = 1, A < B = 0, A = B = 0$
 D. $A > B = 0, A < B = 1, A = B = 1$
- 20) Which of the following logic expressions represents the logic diagram shown?
 A. $F = (A + B)' + CD$
 B. $F = AB' + A'BC$
 C. $F = C'(A + B)' (C + D)$
 D. $F = (A + B)' + C'D$

